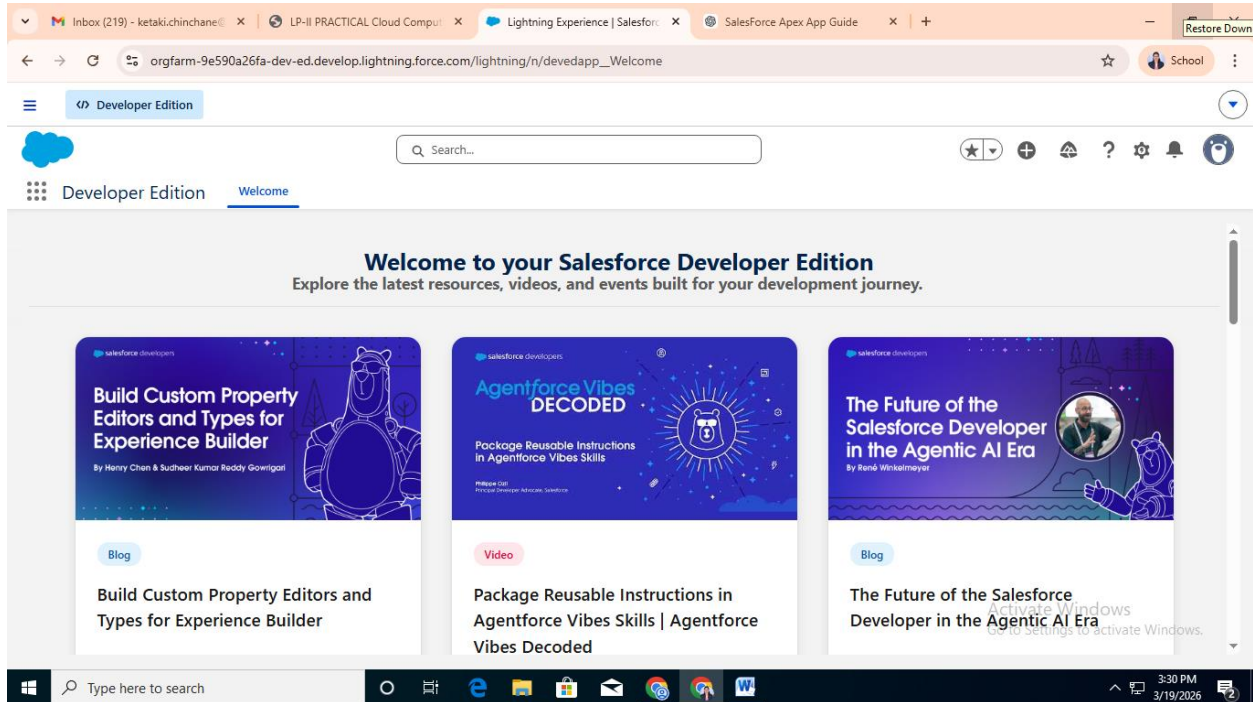


Practical 03: Creating an Application in Salesforce.com using Apex programming Language

Student Management Application

Step 1: Create a Salesforce Account and Login <https://login.salesforce.com/>



Step 2: Create Custom Object (Student)

- Go to **Setup**
- Search: **Object Manager**
- Click **Create** → **Custom Object**
- Fill details:
 - Label: Student
 - Object Name: Student
- Save

Custom Object Definition Edit [Save] [Save & New] [Cancel]

Custom Object Information ! Required Information

The singular and plural labels are used in tabs, page layouts, and reports.

Label Example: Account

Plural Label Example: Accounts

Starts with vowel sound ☐

The Object Name is used when referencing the object via the API.

Object Name Example: Account

Description

Context-Sensitive Help Setting ☒ Open the standard Salesforce.com Help & Training window ☐ Open a window using a Visualforce page

Content Name

Activate Windows
Go to Settings to activate Windows.

SETUP > OBJECT MANAGER

Student_New

Details [Edit] [Delete]

Description

API Name

Custom

✓ Singular Label

Plural Label

Enable Reports

Track Activities

Track Field History

Deployment Status

Help Settings

Standard salesforce.com Help Window

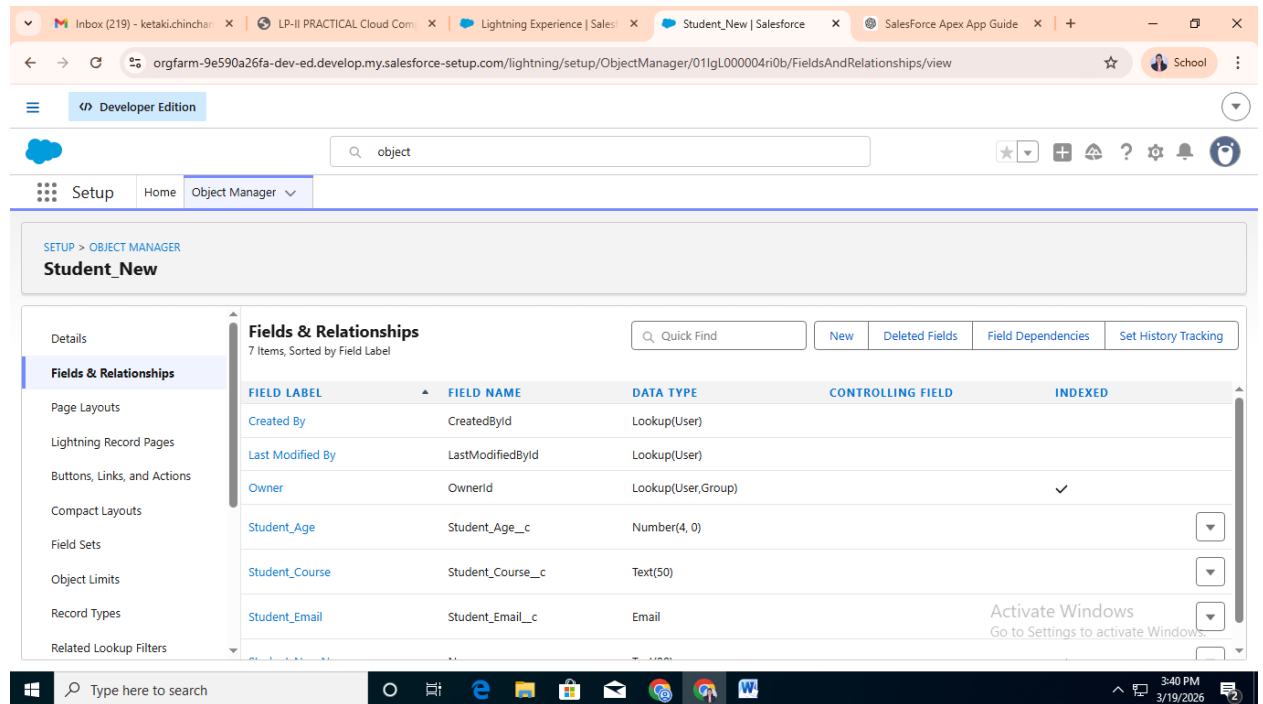
Activate Windows
Go to Settings to activate Windows.

Step 3 – Add Fields:

Go to Fields & Relationship → New → Create all fields below one by one.

- Name (default)
- Age (Number)

- Email (Email)
- Course (Text)



Step 4: Create Apex Class

1. Go to **Setup** → **Apex Classes**
2. Click **New**
3. Write this code:

This class:

- Inserts student data
- Retrieves student records

```
public class StudentController {
```

```
    // Method to insert student
```

```
    public static void addStudent(String name, Integer age, String email, String course) {
```

```
        Student__c s = new Student__c();
```

```

        s.Name = name;

        s.Age__c = age;

        s.Email__c = email;

        s.Course__c = course;

        insert s;

    }

// Method to get all students

public static List<Student__c> getStudents() {

    return [SELECT Id, Name, Age__c, Email__c, Course__c FROM Student__c];

}

}

```

The screenshot displays the Salesforce Developer Edition interface. The browser address bar shows the URL: `orgfarm-9e590a26fa-dev-ed.develop.my.salesforce-setup.com/lightning/setup/ApexClasses/page?address=%2F01pgL00000AKP8F%3FsdclFrameOrigin%3Dhttps%2F...`. The page title is "Apex Classes". The left sidebar contains navigation links such as "Setup Home", "Salesforce Go", "Service Setup Assistant", "Commerce Setup Assistant", "Hyperforce Assistant", "Release Updates", "Salesforce Mobile App", "Lightning Usage", "Sales Cloud Everywhere", and "ADMINISTRATION". The main content area shows the details for the Apex Class "StudentController_New". The class is active, with 0% code coverage. It was created by Ketaki Chinchane on 3/23/2026 at 9:32 PM. The class body is displayed in a code editor, showing the following code:

```

1 public class StudentController_New {
2
3     // Method to insert student
4     public static void addStudent(String name, Integer age, String email, String course) {
5         Student__c s = new Student__c();
6         s.Name = name;
7         s.Age__c = age;
8         s.Email__c = email;
9         s.Course__c = course;
10        insert s;
11    }
12 }

```

The Windows taskbar at the bottom shows the system clock as 3:43 PM on 3/19/2026.

Step 5: Add Validation Rule (Instead of Trigger)

We'll enforce the "Age ≥ 18 " condition using **no-code validation**.

Steps:

1. Go to **Setup** → **Object Manager** → **Student**
2. Click **Validation Rules**
3. Click **New**

Enter:

- **Rule Name:** Age_Validation
- **Formula:**

Age__c < 18

- **Error Message:**
Student must be at least 18 years old.

The screenshot displays the Salesforce Object Manager interface for the 'Student' object. The 'Validation Rules' tab is selected, showing a new validation rule named 'Student_New Validation Rule'. The rule is active and configured with the following details:

Validation Rule Detail	
Rule Name	Age_Validation
Error Condition Formula	Student_Age__c < 18
Error Message	Student must be at least 18 years old.
Description	
Created By	Ketaki Chinchane, 3/23/2026, 9:42 PM
Modified By	Ketaki Chinchane, 3/23/2026, 9:42 PM

The interface includes standard Salesforce navigation elements like the top bar with tabs for Setup, Home, and Object Manager, and a bottom taskbar with various application icons. An 'Activate Windows' watermark is visible in the bottom right corner.

Step 6: Test Your Apex Class

Now test your class from **Developer Console**.

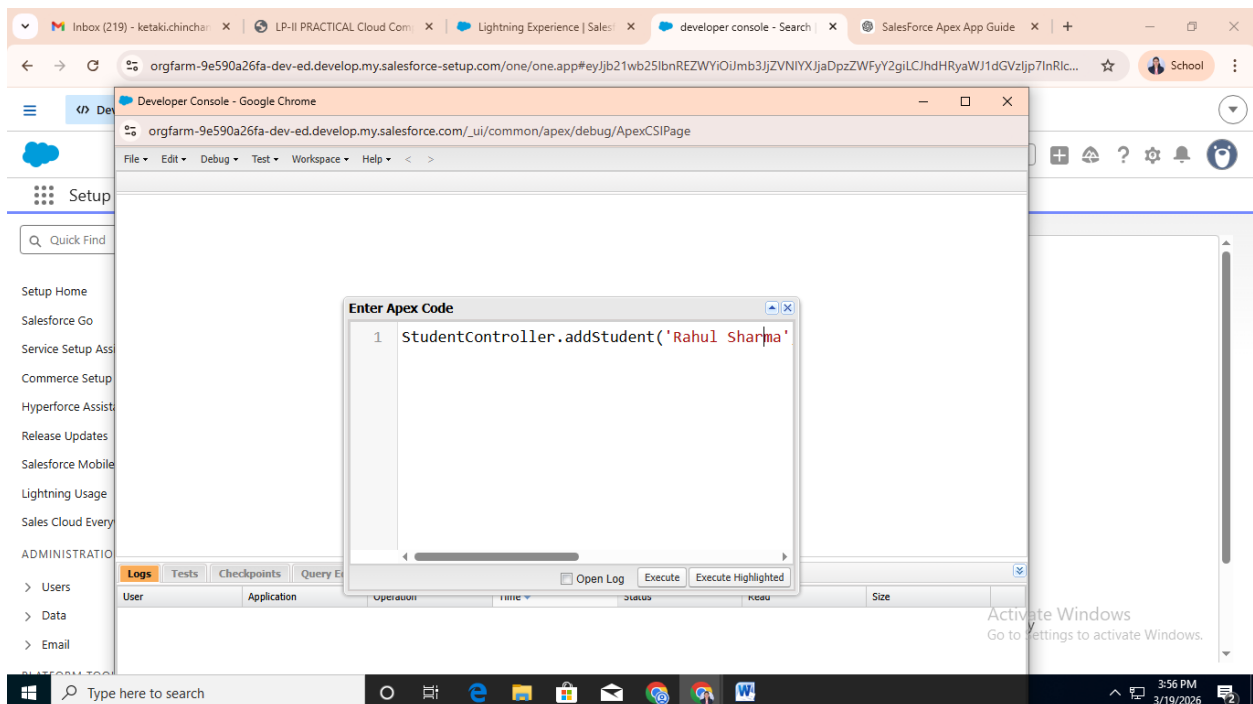
Steps:

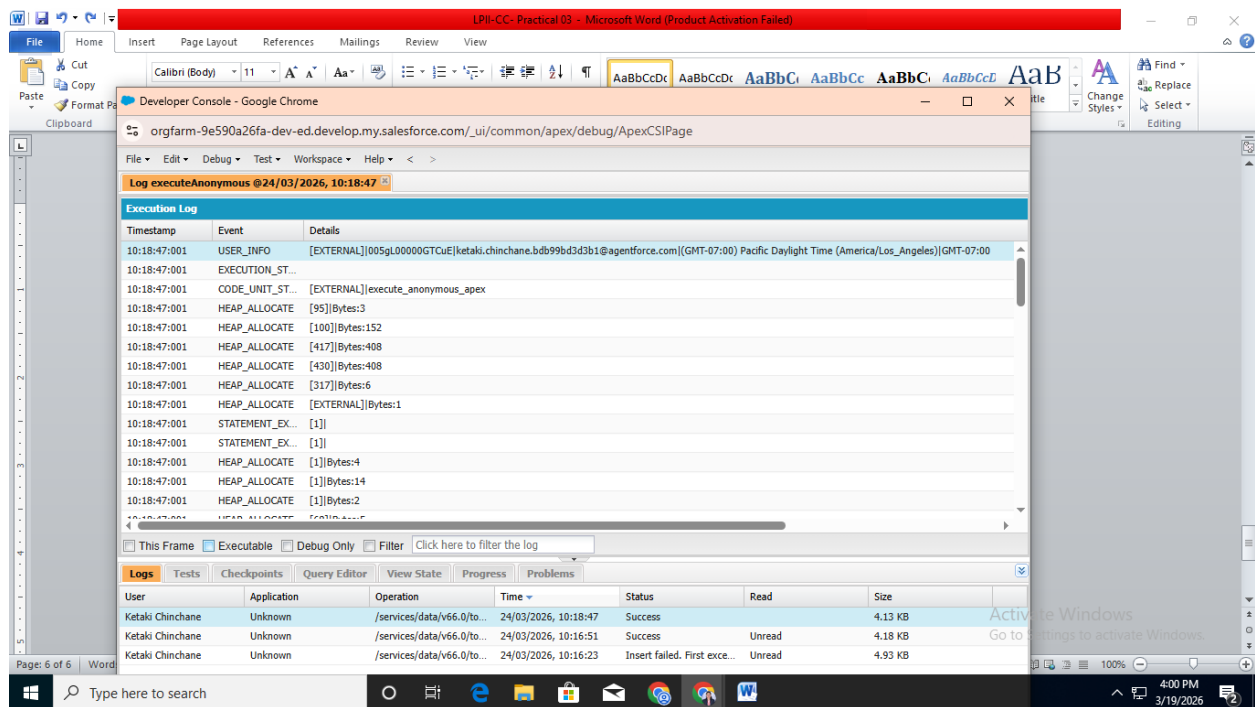
1. Go to setup button but you will find **Developer Console below setup icon.**
2. Go to **Debug → Open Execute Anonymous Window**
3. Run:
4. **Valid case:**

```
StudentController.addStudent('Rahul Sharma', 20, 'rahul@email.com', 'Computer Engineering');
```

Invalid case:

```
StudentController.addStudent('Amit', 16, 'amit@email.com', 'IT');
```





Step 7: Create a Custom App (UI)

Now let's build a simple UI.

Steps:

1. Go to **Setup** → **App Manager**
2. Click **New Lightning App**
3. Enter:
 - App Name: Student App
4. Click **Next** → **Next**
5. Add **Student Object**
6. Finish